The Anomalous Acceleration of the Universe's Expansion

In 1998 two research groups independently discovered an astounding anomaly in the expansion of our universe. For this discovery, the groups' leaders (Saul Perlmutter and Adam Reiss at the University of California, Berkeley, and Brian Schmidt at the Australian National University) won the 2011 Nobel Prize in Physics.

Both groups were observing supernova explosions: explosions triggered when a massive star exhausts its nuclear fuel and implodes to form a neutron star, and the implosion energy blows off the star's outer layers. They discovered that distant supernovae are dimmer than expected, and therefore farther away than expected. Farther enough away that the universe's expansion must have been slower in the past than today. The expansion is accelerating. See Figure 24.4.